

ISHANK AGGARWAL

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EDUCATION

G.L. Bajaj Institute of Technology and Management

2023 – Present

B.Tech in Computer Science and Engineering (Artificial Intelligence)

CGPA: 7.7

Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Databases, Operating Systems

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, SQL

Frontend: React.js, Redux Toolkit, TypeScript, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, JWT Authentication, WebSockets (Socket.IO)

Databases: MongoDB, PostgreSQL, Prisma ORM

AI/ML: OpenAI API, Prompt Engineering, RAG (Basics), Vector Databases

Developer Tools & Platforms: Git, GitHub, Docker, AWS (EC2, S3), Postman, Linux

PROJECTS

Kairo – Distributed Cache Engine

June 2026 – July 2026

Tech Stack: Java 21, HTTP, Docker, Concurrency

[GitHub](#)

- Built a fault-tolerant distributed cache in Java using consistent hashing with 150 virtual nodes per physical node
- Implemented asynchronous replication (RF=2), replica fallback reads, and heartbeat failure detection under 1s
- Designed a pull-based bulk migration protocol enabling zero-downtime rebalancing on node recovery or cluster join
- Validated fault tolerance via a live chaos test: 1,473 requests, 99.79% first-try success, zero hard failures

Devora – Developer Collaboration Platform

May 2026 – June 2026

Tech Stack: React, Redux Toolkit, Node.js, Express, MongoDB, Socket.IO, Docker

[GitHub](#)

- Built a full-stack developer platform unifying DSA practice, coding battles, and team collaboration
- Engineered a custom sandboxed code execution engine using per-submission Docker containers with resource limits
- Developed private coding battles with room-based matchmaking, live leaderboards, and automated scoring

ZyroAI – AI Website Builder

Dec 2025 – Jan 2026

Tech Stack: React, TypeScript, Node.js, PostgreSQL, Prisma, OpenAI API, Stripe

[GitHub](#)

- Integrated OpenAI API with prompt templating, schema validation, and token budget controls per generation
- Implemented automatic code versioning with rollback and pre-render AI output validation
- Built Stripe credit system with per-generation credit deduction and atomic balance management

EXPERIENCE

Freelance Full-Stack Developer

Apr 2025 – June 2025

The Student Adda

Remote

- Led development of a full-stack e-commerce platform as sole engineer in a 3-person team
- Built a role-based admin dashboard for managing products, inventory, and order fulfillment
- Engineered RESTful APIs for product listings, order workflows, and secure admin authentication
- Managed Agile sprints via Jira; owned deployment, post-launch maintenance, and bug resolution in production

ACHIEVEMENTS

Published Patent: [Farmer-Centric Digital Agricultural Advisory System](#), Indian Patent Office (2025)

HackVega 2.0 Finalist – Qualified for the final round among 47,000+ national participants

Solved 750+ Data Structures & Algorithms problems across platforms, tracked on [Codolio](#)

Awarded [Best Full-Stack Developer Certificate](#) for delivering a production-ready web platform

CERTIFICATIONS

• [Java Foundations](#) – JetBrains

• [Agentic AI Foundations](#) – Oracle

• [Docker Foundations](#) – Docker, Inc.

• [System Design](#) – GeeksforGeeks